

UCCN 1004

Data Communications & Networks

Access Control List (ACL)

What is ACL?

- ACL = Access List.
 - essentially a list of conditions that categorize packets.
- It is a form of packet filtering in Cisco router.
- In practical, there are two steps in using ACL:
 - Creating the access list
 - Placing the access list
- Easy to learn, hard to master
- ACL is used in:
 - exercising control over network traffic.
 - filtering unwanted packets when implementing security policies.

“Applying” ACLs

- There are two steps for applying ACL:

- 1st step:

- Create the Access Control List

```
Router(config)#access-list 1 deny 172.16.128.0 0.0.31.255
Router(config)#access-list 1 deny 172.16.48.0 0.0.15.255
Router(config)#access-list 1 deny 172.16.192.0 0.0.63.255
Router(config)#access-list 1 deny 172.16.88.0 0.0.7.255
Router(config)#access-list 1 permit any
```

- 2nd step:

- Place the Access Control List at the Interface

```
Router(config)#interface serial 0
Router(config-if)#ip access-group 1 out
```

- There 3 types of ACL:

- Standard Access List
- Extended Access List
- Named Access List

Creating Standard ACLs

- Command Syntax:

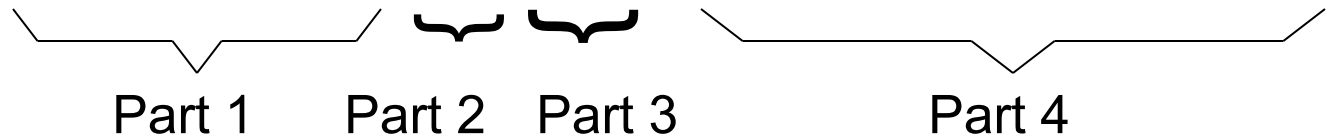
```
R1 (config) #access-list access-list-number {deny |  
permit} {conditions}
```

- Example:

```
R1 (config) #access-list 10 deny host 192.168.1.1
```

```
R1 (config) #access-list 5 permit any
```

```
R1 (config) #access-list 17 deny 172.16.3.0 0.0.0.255
```



- There are 4 parts in the syntax of standard ACL.

General Rule of Standard ACL Syntax

- Part 1:
 - You MUST HAVE the command name “access-list”
- Part 2:
 - Access list number
 - Use any one number from 1 to 99 for standard ACL
- Part 3:
 - Can only be either permit or deny
- Part 4:
 - The “conditions”, view this part as:
 - A host
 - A LAN / IP subnets
 - Any Host
 - Is a source IP or source IP subnets

Syntax of Standard ACLs Explained

- Example:

Part 3:

Can either be permit or deny
(remark is allowed too)

```
R1 (config) #access-list 10 deny host 192.168.1.1
```

Part 1:

You must have
this to create a
standard ACL

Create ACL
in global
configuration
mode

Part 2:

Access list number.
Use 1-99 for
standard ACL.

Part 4:

A host
A LAN / IP subnets
Any Host

More on Syntax of Part 4

Router(config)#access-list 10 permit ?

? = "Part 4" => 3 types of source

host	- A single host address
any	- Any source host
hostname or a.b.c.d	- Address to match

1st type: 1 IP address or 1 source host IP

Router(config)#access-list 10 permit host 192.168.1.1

2nd type: any source IP address

Router(config)#access-list 10 permit any

3rd type: source IP subnet

Router(config)#access-list 10 permit 192.168.1.0 0.0.0.255

Note: Wildcard mask = 0.0.0.255

Introduction to Wildcard Mask

- As you've observed, a portion of "part 4" in defining the IP subnet in ACL contain:
 - A wildcard mask: 0.0.0.255
 - Instead of a subnet mask: 255.255.255.0
- For example: An IP subnet of 172.16.1.0 to 172.16.1.255
 - With subnet mask (in setting static IP route):
 - `R1(config)#ip route 172.16.1.0 255.255.255.0 10.1.1.2`
 - With wildcard mask (in setting access list):
 - `R1(config)#access-list 9 permit 172.16.1.0 0.0.0.255`
- The first step to know wildcard mask is:
 - Wildcard mask IS NOT subnet mask.
 - Wildcard mask DOES NOT HAVE any relation with subnet mask.
 - Wildcard mask use a different set of rules compared to subnet mask.

Applying Wildcard Mask

```
Router(config)#access-list 10 permit 192.168.1.0 0.0.0.255
```

- Step #1: Convert the wildcard mask to binary form:
 - 00000000.00000000.00000000.11111111
- Step #2: Change the '1' to 'x'
 - 00000000.00000000.00000000.xxxxxxxx
- Step #3: Change the IP address to binary:
 - 192.168.1.0 =>
 - 11000000.10101000.00000001.00000000
- Step #4: Match the IP and wildcard mask
 - 11000000.10101000.00000001.00000000
 - 00000000.00000000.00000000.xxxxxxxx
- Algorithm: if wildcard bit = '0', retain IP bits, if wildcard bit = 'x', replace.
- Results:
 - 11000000.10101000.00000001.xxxxxxxx
- In the place of 'x', it can be either '0' or '1'.

How Wildcard Mask Works?

```
R1 (config)#access-list 2 deny 172.16.0.1 0.0.255.254
```

- Question:
 - In the above statement, what IPs does the ACL deny?

- Answer

Access-list 1 permit 172.16.0.1 0.0.255.254

	172	16	0	1	0	0	255	254
IP Address	1	0	1	0	1	1	0	0
Wildcard mask	0	0	0	0	0	0	0	0
Match Value	1	0	1	0	1	1	0	0

#Quick Quiz#

R1 (config) #access-list 2 permit 172.16.1.3 0.0.0.252

- Question:
 - Given above statement, does the following IP be “permitted” by the access list?
 - A) 172.16.1.5
 - B) 172.16.1.51
 - C) 172.16.1.13

#Quick Quiz#

```
R1 (config)#access-list 2 permit 172.16.1.3  
255.255.255.252
```

- Question:
 - Given above statement, (accidentally mistaken subnet mask as wildcard mask) does the following IP be “permitted” by the access list?
 - A) 172.16.1.5
 - B) 172.16.1.51
 - C) 172.16.1.13

The ANY and HOST Option

```
R1 (config) #access-list 4 permit 0.0.0.0 255.255.255.255
```

Can be written as:

```
R1 (config) #access-list 4 permit any
```

```
R1 (config) #access-list 3 permit 172.16.3.2 0.0.0.0
```

Can be written as:

```
R1 (config) #access-list 3 permit host 172.16.3.2
```

- There are two special keywords that are used in ACLs, the any and host options.
- Simply put, the any option substitutes 0.0.0.0 for the IP address and 255.255.255.255 for the wildcard mask.
- This option will match any address that it is compared against.
- The host option substitutes for the 0.0.0.0 mask.
- This mask requires that all bits of the ACL address and the packet address match. This option will match just one address.

More Rules for Standard ACLs

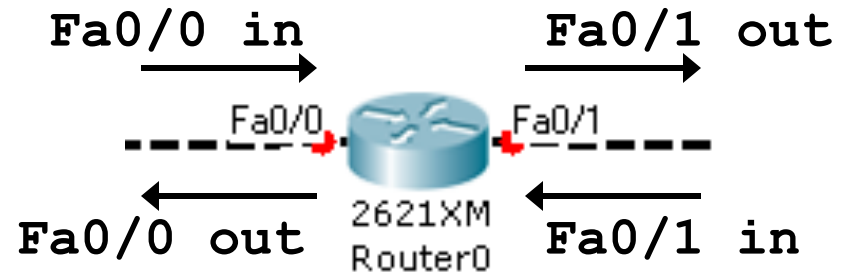
- The “part 4” of Standard ACL works on the source IP address.
- 1 ACL list represent by 1 ACL number.
- You can assign as many “conditions” into a list (e.g. next slide)
- Every Standard ACL list has a hidden “deny all” statement at the last of the list.
- Use this command to see the access list
 - **Router#show access-list**
 - **Router#show access-list 5**
- Use this command to erase a access list
 - **Router(config)#no access-list 15**

Placing Standard ACLs - 1

- Syntax:

```
R1 (config) #interface {interface-name}
```

```
R1 (config-if) # {protocol} access-group access-  
list-number {in | out}
```



- Example:

```
R1 (config) #int fa0/0
```

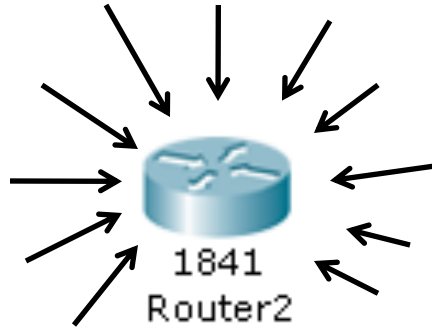
```
R1 (config-if) #ip access-group 10 in
```

```
R1 (config-if) #ip access-group 12 out
```

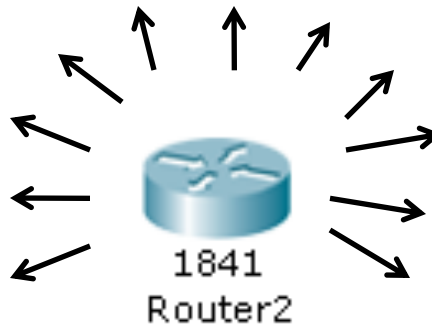
- The **ip access-group** command links an existing ACL to an interface

Placing Standard ACLs - 2

Direction:
IN



Direction:
OUT



Placing Standard ACLs - 3

- You can place

- same Access List at different interface

```
Router(config)#int fa0/0
Router(config-if)#ip access-group 10 in
Router(config)#int fa0/1
Router(config-if)#ip access-group 10 in
```

- only 2 Access List at same interface for both in and out

```
Router(config)#int fa0/0
Router(config-if)#ip access-group 10 in
Router(config-if)#ip access-group 97 out
```

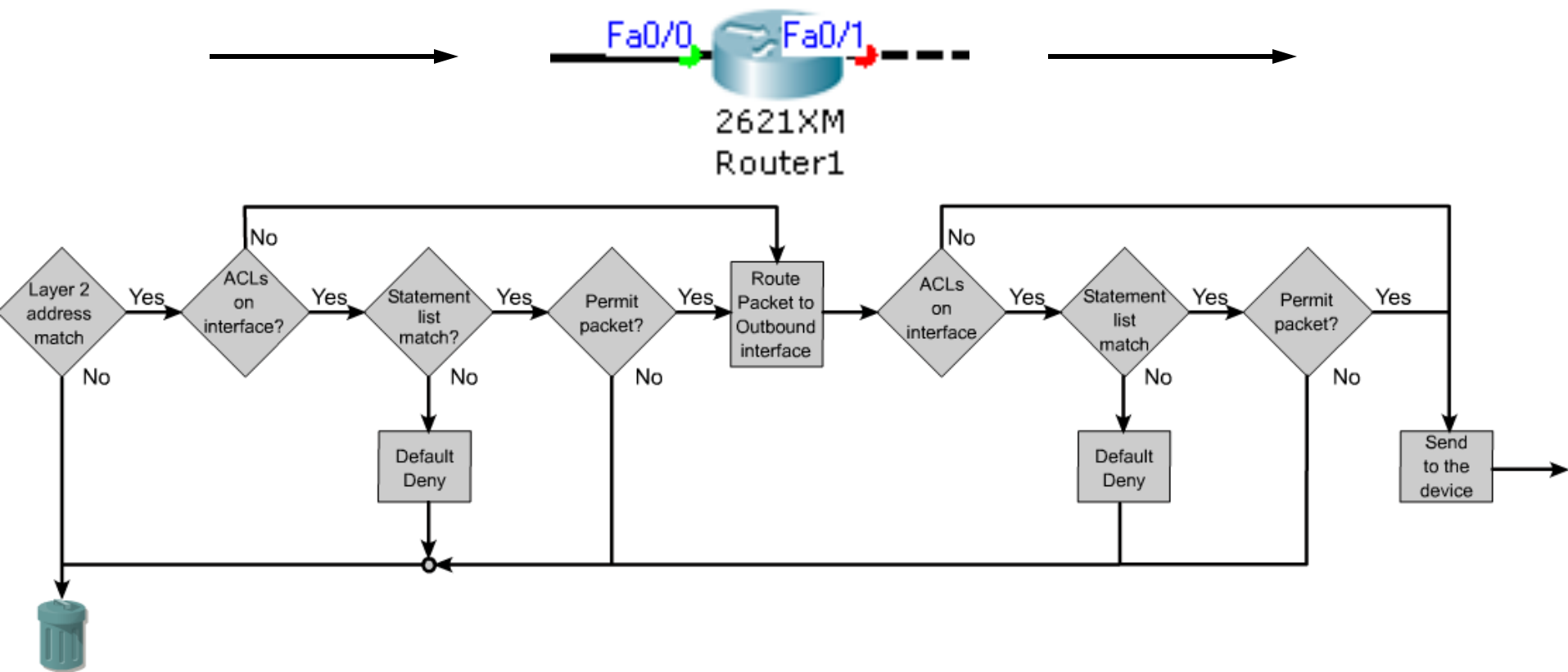
- ACL of different layer 3 protocols at same interface (in and out)

```
Router(config)#int fa0/0
Router(config-if)#ip access-group 10 in
Router(config-if)#ip access-group 97 out
Router(config-if)#ipx....
```

ACL and Routing Processes in a Router

ip access-group *list_number* in

ip access-group *list_number* out



Checking and Unbind the “Placement” of ACLs

- Use the following commands to verify the placement of ACLs on the interfaces of router.
 - Router#show ip interface
 - Router#show running-config
- Remember to “unbind” an ACL at the interface level by typing the following commands

```
R1(config)#int fa0/0
```

```
R1(config-if)#no ip access-group list_number in
```

Show IP Interface

```
R1#show ip interface
FastEthernet0/0 is up, line protocol is up (connected)
  Internet address is 192.168.14.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Multicast reserved groups joined: 224.0.0.9
Outgoing access list is not set
Inbound access list is 1
  Proxy ARP is enabled
  Security level is default
  Split horizon is enabled
```

...

- The show ip interface command displays IP interface information and indicates whether any ACLs are set.

While you delete Access List

- When you delete a particular ACLs,
 - Only the content of the ACL is empty
 - The binding ACLs is still at the interface
 - Meaning the Interface still has the ACL
 - Problem occurs if you “unintentionally” re-use the deleted ACL...

Example:

```
R1(config)#no access-list 2
R1(config)#exit
R1#show run
version 12.2
hostname R1
interface FastEthernet0/0
  ip address 192.168.14.1 255.255.255.0
  ip access-group 2 in
!
```

Placing ACL at VTY

- Here is an example of allowing only host 172.16.10.3 to telnet into a router:

```
Lab_A(config)#access-list 50 permit host 172.16.10.3
Lab_A(config)#line vty 0 4
Lab_A(config-line)#access-class 50 in
```

- Because of the implied deny any at the end of the list, the access list stops any host from telnetting into the router except the host 172.16.10.3, regardless of which individual IP address on the router is used as a target.
- For VTY line, use **access-class** to bind the ACL
 - instead of **ip access-group** for other type of interfaces.

Now: Syntax for Extended ACL

```
Router(config)#access-list access-list-number {permit | deny}  
protocol source  
[source-mask destination destination-mask operator operand]  
[established]
```

Paramter	Description
<i>access-list-number</i>	Identifies the list using a number in the range 100 to 199.
permit deny	Indicates whether this entry allows or blocks the specified address.
<i>protocol</i>	The protocol, such as IP, TCP, UDP, ICMP, GRE, or IGRP.
source and destination	Identifies source and destination addresses.
<i>source-mask</i> and <i>destination-mask</i>	Wildcard mask; zeros indicate positions that must match, ones indicate do not care positions.
<i>operator operand</i>	lt, gt, eq, neq (less than, greater than, equal, not equal), and a port number.
established	Allows TCP traffic to pass if the packet uses an established connection (for example, has ACK bits set).

Creating IP Extended List

```
R1(config)#access-list 114 permit tcp 172.16.6.0 0.0.0.255 any eq telnet
R1(config)#access-list 114 permit tcp 172.16.6.0 0.0.0.255 any eq ftp
R1(config)#access-list 114 permit tcp 172.16.6.0 0.0.0.255 any eq ftp-data
```

Part 1 Part 2 Part 3 Part 4 Part 5 Part 6 Part 7 Part 8

- In Extended Access List

- We have both source IP and destination IP
- We have layer-4 protocols and layer-3 protocols
- We have a different access-list number range
- Instead of “4 parts” in standard ACL, we have 8 parts in extended ACL

Extended ACL – Part 1 & 2

- Part 1: The command “access-list” is still the same.
- Part 2: The access-list number range for extended ACL is 100 to 199, and 2000 to 2699
 - You must match the ACL number according to the following guideline!!

Lab_A(config)#access-list ?

<1-99>	IP standard access list
<100-199>	IP extended access list
<1000-1099>	IPX SAP access list
<1100-1199>	Extended 48-bit MAC address access list
<1200-1299>	IPX summary address access list
<1300-1999>	IP standard access list (expanded range)
<200-299>	Protocol type-code access list
<2000-2699>	IP extended access list (expanded range)
<300-399>	DECnet access list
<600-699>	Appletalk access list
<700-799>	48-bit MAC address access list
<800-899>	IPX standard access list
<900-999>	IPX extended access list

Extended ACL – Part 3 & 4

```
Lab_A(config)#access-list 110 ?
```

deny	Specify packet
dynamic	Specify a DYNAMIC list of PERMITs or DENYs
permit	Specify packets to forward

- Part 3:
 - deny or permit
- Part 4:
 - Layer 3 protocol
 - Layer 4 protocol

```
Lab_A(config)#access-list 110 deny ?
```

<0-255>	An IP protocol number
eigrp	Cisco's EIGRP routing protocol
gre	Cisco's GRE tunneling
icmp	Internet Control Message Protocol
igmp	Internet Gateway Message Protocol
igrp	Cisco's IGRP routing protocol
ip	Any Internet Protocol
ipinip	IP in IP tunneling
nos	KA9Q NOS compatible IP over IP tunneling
ospf	OSPF routing protocol
tcp	Transmission Control Protocol
udp	User Datagram Protocol

Extended ACL – Part 5 and 6

- Part 5:

- Source address

- Host, any, IP subnet
 - Same as standard ACL

```
Lab_A(config)#access-list 110 deny tcp ?
```

```
A.B.C.D Source address
```

```
any Any source host
```

```
host A single source host
```

- Part 6:

- “Normally” Destination address

```
Lab_A(config)#access-list 110 deny tcp any ?
```

```
A.B.C.D Destination address
```

```
any Any destination host
```

```
eq Match only packets on a given port number
```

```
gt Match only packets with a greater port number
```

```
host A single destination host
```

```
lt Match only packets with a lower port number
```

```
neq Match only packets not on a given port number
```

```
range Match only packets in the range of port numbers
```

Extended ACL – Part 7

- Part 7

- Normally conditional phrase

- gt = greater than
 - eq = equal
 - lt = less than
 - neq = not equal

```
Lab_A(config)#access-list 110 deny tcp any host 172.16.30.2 ?
```

eq	Match only packets on a given port number
established	Match established connections
fragments	Check fragments
gt	Match only packets with a greater port number
log	Log matches against this entry
log-input	Log matches against this entry, including input interface
lt	Match only packets with a lower port number
neq	Match only packets not on a given port number
precedence	Match packets with given precedence value
range	Match only packets in the range of port numbers
tos	Match packets with given TOS value

Checking Your Extended ACL

- Show access-list commands displays the contents of all ACLs on the router.

```
Router#show access-lists
Standard IP access list 2
  deny   172.16.1.1
  permit 172.16.1.0, wildcard bits 0.0.0.255
  deny   172.16.0.0, wildcard bits 0.0.255.255
  permit 172.0.0.0, wildcard bits 0.255.255.255
Extended IP access list 101
  permit tcp 192.168.6.0 0.0.0.255 any eq telnet
  permit tcp 192.168.6.0 0.0.0.255 any eq ftp
  permit tcp 192.168.0.0 0.0.0.255 any eq ftp-data
Router#
```

Output from the **show access-lists** command showing the two access lists that have been configured.

Extended ACL – Part 8

- Part 8:
 - Port numbers
 - Port number equivalent network service
 - telnet (port 22)

```
Lab_A(config)#access-list 110 deny tcp any host 172.16.30.2 eq ?
```

<0-65535>	Port number
bgp	Border Gateway Protocol (179)
chargen	Character generator (19)
cmd	Remote commands (rcmd, 514)
daytime	Daytime (13)
discard	Discard (9)
domain	Domain Name Service (53)
echo	Echo (7)
exec	Exec (rsh, 512)
finger	Finger (79)
ftp	File Transfer Protocol (21)
ftp-data	FTP data connections (20, 21)
gopher	Gopher (70)
hostname	NIC hostname server (101)
ident	Ident Protocol (113)
irc	Internet Relay Chat (194)
klogin	Kerberos login (543)
ksell	Kerberos shell (544)
login	Login (rlogin, 513)
lpd	Printer service (515)
nntp	Network News Transport Protocol (119)
pim-auto-RP	PIM Auto-RP
pop2	Post Office Protocol v2 (109)
pop3	Post Office Protocol v3 (110)
smtp	Simple Mail Transport Protocol (25)
sunrpc	Sun Remote Procedure Call (111)
syslog	Syslog (514)
tacacs	TAC Access Control System (49)
talk	Talk (517)
telnet	Telnet (23)
time	Time (37)
uucp	Unix-to-Unix Copy Program (540)
whois	Nickname (43)
www	World Wide Web (HTTP, 80)

Extended List Placement

```
R1 (config-if) #ip access-group 110 in  
R1 (config-if) #ip access-group 110 out
```

```
R1#show ip interface  
R1#show run
```

- Same rules applied as standard ACL.
 - For placement or assigning
 - For unbinding
 - For checking placement

Creating Named ACLs - 1

- A named ACL is created with the ip access-list command.
- The advantages that a named access list provides are:
 - Intuitively identify an ACL using an alphanumeric name.
 - Eliminate the limit of 798 simple and 799 extended ACLs
- Named ACLs provide the ability to modify ACLs without deleting and then reconfiguring them.
- It is important to note that a named access list will allow the deletion of statements but numbered ACL will only allow for statements to be inserted at the end of a list. Even with named ACLs it is a good idea to use a text editor to create them.
- Consider the following before implementing named ACLs.
 - Named ACLs are not compatible with Cisco IOS releases prior to Release 11.2.
 - The same name may not be used for multiple ACLs. For example, it is not permissible to specify both a standard and extended ACL named George.

Creating Named ACLs (Standard)

- IP named ACLs were introduced in Cisco IOS Software Release 11.2, allowing standard and extended ACLs to be given names instead of numbers.

```
Lab_A#config t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
Lab_A(config)#ip access-list ?
```

```
extended  Extended Acc
logging   Control access list logging
standard  Standard Access List
```

```
Lab_A(config)#ip access-list standard ?
```

```
<1-99>  Standard IP access-list number
WORD    Access-list name
```

```
Lab_A(config)#ip access-list standard BlockSales
```

```
Lab_A(config-std-nacl)#
```

continue

```
Lab_A(config-std-nacl)#?
```

```
Standard Access List configuration commands:
```

```
default  Set a command to its defaults
deny     Specify packets to reject
exit     Exit from access-list configuration mode
no       Negate a command or set its defaults
permit   Specify packets to forward
```

```
Lab_A(config-std-nacl)#deny 172.16.40.0 0.0.0.255
```

```
Lab_A(config-std-nacl)#permit any
```

```
Lab_A(config-std-nacl)#exit
```

```
Lab_A(config)#^Z
```

```
Lab_A#
```

Placing Name ACLs

```
Lab_A#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Lab_A(config)#int e1
Lab_A(config-if)#ip access-group BlockSales out
Lab_A(config-if)#^Z
Lab_A#
```

- Same as standard ACL and extended ACL
- Using **ip access-group** command

Creating Named ACLs – Extended -1

```
router#configure terminal
Enter configuration commands, one per line.
router(config)#ip access-list extended test
router(config-ext-nacl)#permit ip host 2.2.2.2 host
3.3.3.3
router(config-ext-nacl)#permit tcp host 1.1.1.1 host
5.5.5.5 eq www
router(config-ext-nacl)#permit icmp any any
router(config-ext-nacl)#permit udp host 6.6.6.6 10.10.10.0
0.0.0.255 eq domain
router(config-ext-nacl)#^Z
1d00h: %SYS-5-CONFIG_I: Configured from console by
consoles-l
router#show access-list
Extended IP access list test
    permit ip host 2.2.2.2 host 3.3.3.3
    permit tcp host 1.1.1.1 host 5.5.5.5 eq www
    permit icmp any any
    permit udp host 6.6.6.6 10.10.10.0 0.0.0.255 eq
    domain
```

Creating Named ACLs – Extended -2

```
router(config)#ip access-list extended test
!--- ACL entry deleted.
router(config-ext-nacl)#no permit icmp any any
!--- ACL entry added.
router(config-ext-nacl)#permit gre host 4.4.4.4 host
      8.8.8.8
router(config-ext-nacl)#^Z
1d00h: %SYS-5-CONFIG_I: Configured from console by
      consoles-1
!--- ACL entry added.
router(config-ext-nacl)#permit gre host 4.4.4.4 host
      8.8.8.8
router(config-ext-nacl)#^Z
1d00h: %SYS-5-CONFIG_I: Configured from console by
      consoles-1

router#show access-list
Extended IP access list test
    permit ip host 2.2.2.2 host 3.3.3.3
    permit tcp host 1.1.1.1 host 5.5.5.5 eq www
    permit udp host 6.6.6.6 10.10.10.0 0.0.0.255 eq
    domain
```

Inserting a line in Named ACL - 1

Router(config)# ip access-list extended serveraccess2	Creates an extended named ACL called serveraccess2.
Router(config-ext-nacl)# 10 permit tcp any host 131.108.101.99 eq smtp	Uses a sequence number 10 for this line.
Router(config-ext-nacl)# 20 permit udp any host 131.108.101.99 eq domain	Sequence number 20 will be applied after line 10.
Router(config-ext-nacl)# 30 deny ip any any log	Sequence number 30 will be applied after line 20.

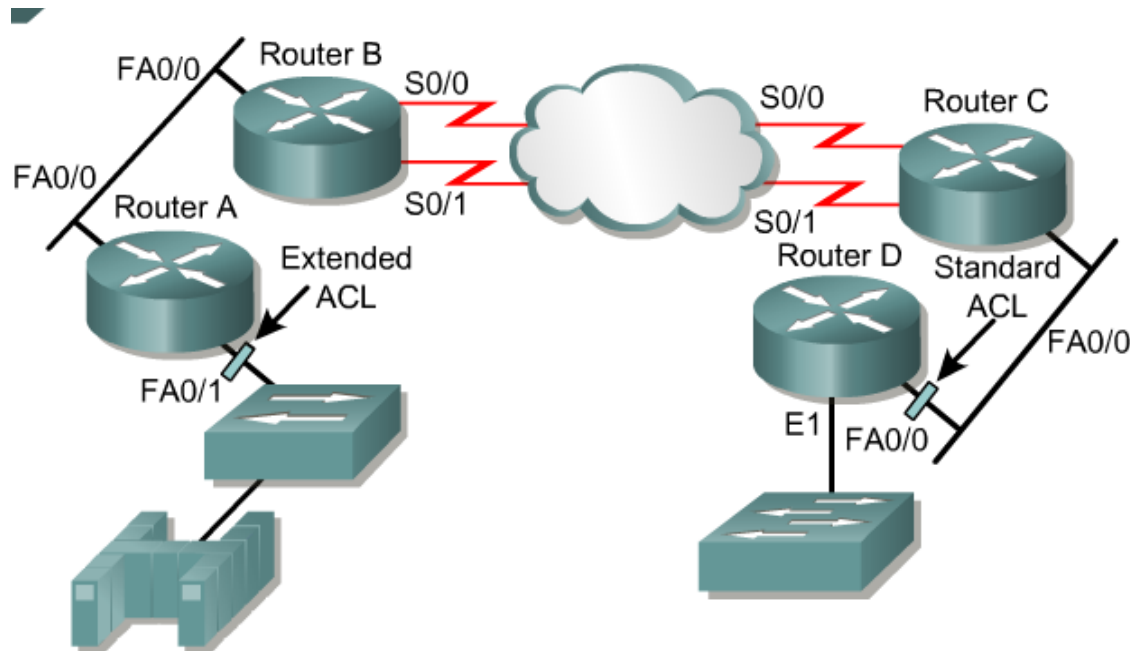
Inserting a line in Named ACL - 2

Router(config-ext-nacl)# exit	Returns to global configuration mode.
Router(config)# interface fastethernet 0/0	Moves to interface configuration mode.
Router(config-if)# ip access-group serveraccess2 out	Applies this ACL in an outbound direction.
Router(config-if)# exit	Returns to global configuration mode.
Router(config)# ip access-list extended serveraccess2	Moves to named ACL configuration mode for the ACL serveraccess2.
Router(config-ext-nacl)# 25 permit tcp any host 131.108.101.99 eq ftp	Sequence number 25 places this line after line 20 and before line 30.
Router(config-ext-nacl)# exit	Returns to global configuration mode.

Summary: Monitoring ACL

Command	Effect
<code>show access-list</code>	Displays all access lists and their parameters configured on the router. This command does not show you which interface the list is set on.
<code>show access-list 110</code>	Shows only the parameters for the access list 110. This command does not show you the interface the list is set on.
<code>show ip access-list</code>	Shows only the IP access lists configured on the router.
<code>show ip interface</code>	Shows which interfaces have access lists set.
<code>show running-config</code>	Shows the access lists and which interfaces have access lists set.

Summary: Placing ACLs



- The general rule is to put the extended ACLs as close as possible to the source of the traffic denied.
- Standard ACLs do not specify destination addresses, so they should be placed as close to the destination as possible.